

ReDim: Quantifying dependence via dimension reduction

Applicant:

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1 Scientific aspects

1.1 State of the art and preliminary work

Detecting statistical association among several random variables is an ubiquitous task. Measures of association capture the many facets of dependence relationships - from classical linear correlation coefficients to indices for detecting monotone associations, tail dependence, asymmetric dependence etc. Most concepts of statistical association, however, consider association as an undirected property, i.e., the association among the variables remains unchanged when permuting the variables. This includes Pearson's correlation coefficient, measures of bivariate and multivariate concordance like Kendall's τ , Spearman's ρ_S or Gini's γ (see, e.g., [6, 10, 18, 30, 37, 59, 70]), distance covariance and distance multivariate (see, e.g., [4, 69]), the maximal information coefficient (see, e.g., [58]), the randomized dependence coefficient (see, e.g., [51]), coefficients of tail dependence (see, e.g., [38]), and various measures of asymmetry (see, e.g., [36, 43, 55]).

In contrast, when causality is present, one variable may have a stronger influence on another variable than vice versa. This is why quantifying the degree of predictability or explainability of a single (continuous) random variable Y using the information contained in a set of potential explanatory (continuous) random variables $X_1, \dots, X_d$, $d \geq 1$, requires measures of directed association. An index measuring such a degree should be capable of detecting perfect dependence also known as complete dependence (see, e.g., [34, 49, 71]): Y is said to be *completely dependent* on $\mathbf{X} = (X_1, \dots, X_d)$ if there exists a measurable function f such that

$$Y = f(\mathbf{X}) \quad \text{almost surely.} \quad (1)$$

Regardless of whether one is interested in predictability or explainability with respect to a single or a set of explanatory variables, the concept of complete dependence - being the exact opposite of independence - is essential in many ways:

Predictability: Measures of directed dependence of a random variable Y on a random vector $\mathbf{X}$.

In Bernoulli News 2021, Han [35] pointed out the need for dependence measures μ being *scale-invariant* (i.e., independent on the marginals) and fulfilling the following properties:

- (P.1) $\mu \in [0, 1]$,
- (P.2) $\mu = 0$ if and only if Y and $\mathbf{X}$ are independent,
- (P.3) $\mu = 1$ if and only if Y is completely dependent on $\mathbf{X}$.

Quite recently, Azadkia and Chatterjee [3] introduced their so-called 'simple measure of conditional

dependence' which is based on Chatterjee [5], Dette et al. [8], Siburg and Stoimenov [64] and, for a $(d + 1)$ -dimensional random vector $(\mathbf{X}, Y)$, given by

$$T(Y|\mathbf{X}) := \frac{\int_{\mathbb{R}} \text{var}(P(Y \geq y | \mathbf{X})) \, dP^Y(y)}{\int_{\mathbb{R}} \text{var}(\mathbb{1}_{(-\infty, y]}(Y)) \, dP^Y(y)} \quad (2)$$

Also recently, Griessenberger et al. [34] introduced their index $\zeta_1 := \zeta_1(Y|\mathbf{X})$ based on the Linkage transformation and a related metric as used for T . Both the maps T and ζ_1 quantify the scale-invariant (rank-based) extent of dependence of the random variable Y on the random vector $\mathbf{X}$, and belong to a class of indices fulfilling (P.1) - (P.3) that also includes [17, 40, 67, 71, 75].

Measures of directed dependence are of high practical relevance in all fields of application and research where predictability plays a central role. This includes biology [5, 67], climatology [20, 40], economics [40] etc.; further real data analyses may be found in [3].

Explainability: Scale-invariant coefficient of determination - explaining variance:

Similar to the coefficient of determination in regression analysis, the Sobol index

$$\frac{\text{var}(E(Y | \mathbf{X}))}{\text{var}(Y)}$$

(see, e.g., [27, 68, 73, 74]; also known as correlation ratio [63]) measures the proportion of the variance of $Y \in L^2$ explained by the regression function $r(\mathbf{x}) := E(Y|\mathbf{X} = \mathbf{x})$. Denoting by F_i the marginal distribution function of X_i , $i \in \{1, \dots, d\}$, and by G the marginal distribution function of Y and setting $\mathbf{F}(\mathbf{X}) := (F_1(X_1), \dots, F_d(X_d))$, the scale-invariant correlation ratio (in the following referred to as *scale-invariant coefficient of determination*) is given by

$$R^2(Y|\mathbf{X}) := \frac{\text{var}(E(G(Y) | \mathbf{F}(\mathbf{X})))}{\text{var}(G(Y))} \quad (3)$$

The scale-invariant R^2 fulfills (P.1) and (P.3) but fails to satisfy (P.2), in general. However, a weak version of (P.2) is obtained: $R^2(Y|\mathbf{X}) = 0$ whenever Y and $\mathbf{X}$ are independent.

Like measures of directed dependence, the coefficient of determination appears in all quantitatively working disciplines, and there is a great need for fast, precise and efficient estimators for these quantities.

Dimension reduction: When investigating scale-invariant (rank-based) mappings for random vectors with continuous marginals it is natural to work with copulas (we denote by $\mathcal{C}^{d+1}$ the family of all $(d + 1)$ -dimensional copulas) which, by Sklar's theorem (see, e.g., [12, 54]), capture all scale-invariant dependence structures between random variables: For every random vector $(\mathbf{X}, Y)$ with continuous marginals there exists

a unique copula $A \in \mathcal{C}^{d+1}$ such that

$$P(X_1 \leq x_1, \dots, X_d \leq x_d, Y \leq y) = A(F_1(x_1), \dots, F_d(x_d), G(y)) \quad (4)$$

for all $(x_1, \dots, x_d, y) \in \mathbb{R}^{d+1}$. The probability integral transform $(\mathbf{F}(\mathbf{X}), G(Y))$ then has distribution function A , and the values $T(Y|\mathbf{X})$ and $R^2(Y|\mathbf{X})$, respectively, possess alternative representations in terms of the well-known measures of bivariate association Spearman's footrule and Spearman's ρ_S , respectively, both of which evaluate a certain bivariate copula $\psi(A)$ derived from A via the map $\psi : \mathcal{C}^{d+1} \rightarrow \mathcal{C}^2$ given by

$$\psi(A)(s, t) := \int_{[0,1]^d} K_A(\mathbf{u}, [0, s]) K_A(\mathbf{u}, [0, t]) \, d\mu_{A^{1:d}}(\mathbf{u}) \quad (5)$$

for all $(s, t) \in [0, 1]^2$ (see [20]). Here, $K_A : [0, 1]^d \times \mathcal{B}([0, 1]) \rightarrow [0, 1]$ denotes the Markov kernel¹ of A from $[0, 1]^d$ to the Borel σ -field $\mathcal{B}([0, 1])$ on $[0, 1]$, and $A^{1:d}$ denotes the d -dimensional marginal copula of A with respect to the first d coordinates. In what follows we denote by $\Psi := \psi(\mathcal{C}^{d+1}) \subseteq \mathcal{C}^2$ the set of all bivariate copulas emerging from the transformation ψ .

The map ψ transforms every $(d + 1)$ -dimensional copula to an exchangeable bivariate copula (see Figures 1, 2 and 3 for an illustration).

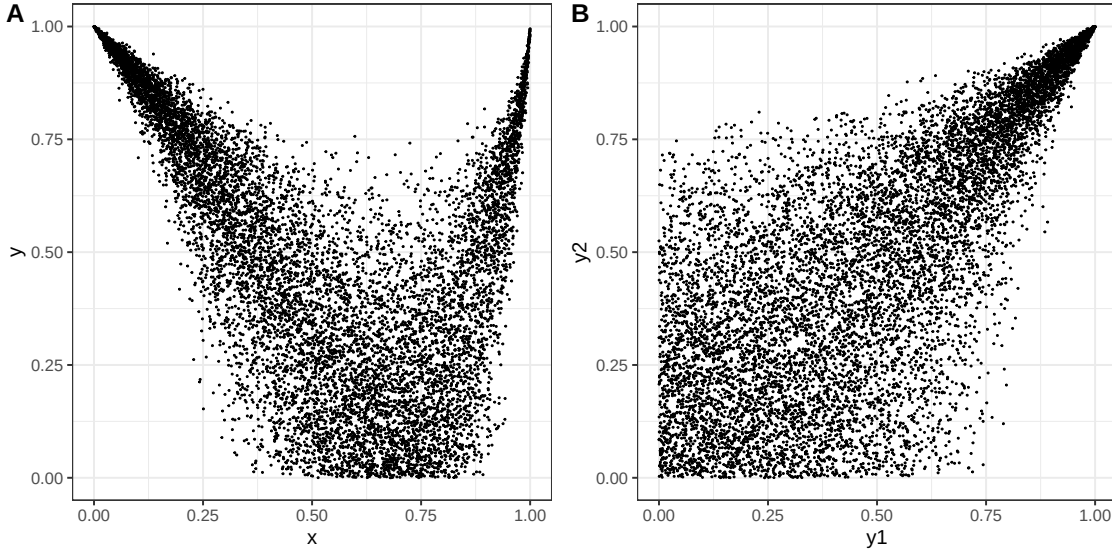


Figure 1: Sample of size $n = 10,000$ drawn from a copula A linking a standard normal random variable X and random variable $Y = 2X^2 + 2X + \varepsilon$ with standard normal noise ε (left panel), together with a sample of equal size from $\psi(A)$ (right panel).

¹For more information on Markov kernels, also in the context of copulas, we refer to [12, 42, 48, 47, 53]

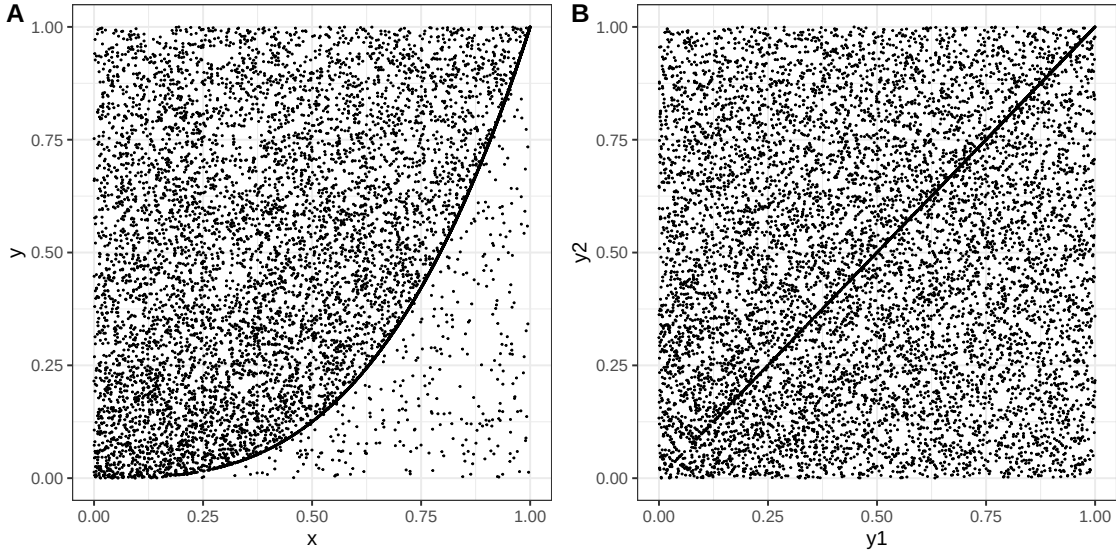


Figure 2: Sample of size $n = 10.000$ drawn from a Marshall-Olkin copula A (see [12, Example 6.6.8]) with parameter $\alpha = 0.9$ and $\beta = 0.3$ (left panel), together with a sample of equal size from $\psi(A)$ (right panel).

Although $\psi(A)$ is 2-dimensional, it nevertheless contains the full information on whether Y is completely dependent on the d -dimensional random vector $\mathbf{X}$, and whether Y and $\mathbf{X}$ are independent (see [20] and Theorem 1.1 below); as usual, we denote by M the comonotonicity copula and Π denotes the independence copula (see, e.g., [12, 54]):

1.1 Theorem.

1. $\psi(A) = M$ if and only if Y is completely dependent on the d -dimensional random vector $\mathbf{X}$.
2. $\psi(A) = \Pi$ if and only if Y and the d -dimensional random vector $\mathbf{X}$ are independent.

The transformation ψ can be seen as an aggregation/dimension reduction containing all relevant information, and is therefore the key object of this proposal.

The 2-dimensional copula $\psi(A)$ itself can be interpreted as the distribution function of two random variables $V := G(Y)$ and $V' := G(Y')$ with $(\mathbf{X}, Y)$ and $(\mathbf{X}, Y')$ sharing the same distribution and being conditionally independent given $\mathbf{X}$, i.e.,

$$\psi(A)(s, t) = P(V \leq s, V' \leq t)$$

for all $(s, t) \in [0, 1]^2$. Moreover, the copula $\psi(A)$ may also be regarded as a generalization of the well-known bivariate Markov product of copulas: For two copulas $B, C \in \mathcal{C}^2$ the *Markov product* $B * C : [0, 1]^2 \rightarrow [0, 1]$

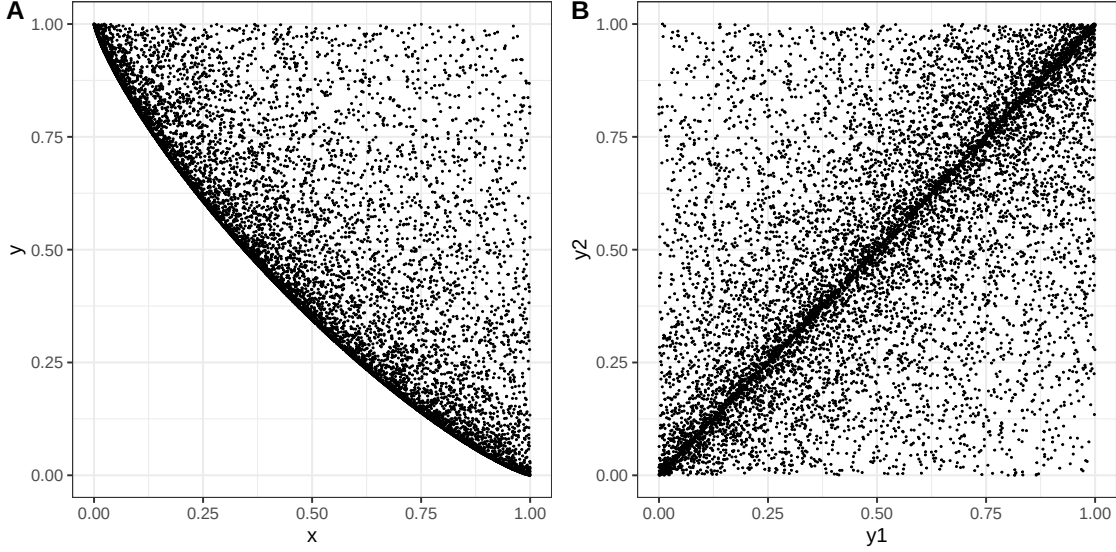


Figure 3: Sample of size $n = 10.000$ drawn from a Clayton copula A (see [12, Example 6.5.17]) with parameter $\alpha = -0.75$ (left panel), together with a sample of equal size from $\psi(A)$ (right panel).

given by

$$(B * C)(s, t) := \int_{[0,1]} K_{B^t}(u, [0, s]) K_C(u, [0, t]) d\lambda(u)$$

is well-known to be a copula (see [7, 72] and [12, Chapter 5]); here, B^t denotes the transpose of B . In the bivariate case, i.e., for $d = 1$, we therefore have $\psi(A) = A^t * A$.

Predictability: Measures of directed dependence of a random variable Y on a random vector $\mathbf{X}$.

The index T fulfills

$$T(Y|\mathbf{X}) = \varphi(\psi(A)) \quad (6)$$

(see [20]) where $\varphi : \mathcal{C}^2 \rightarrow \mathbb{R}$ denotes Spearman's footrule² given by

$$\varphi(C) := 6 \int_{[0,1]} C(t, t) d\lambda(t) - 2$$

and λ the Lebesgue measure on $\mathcal{B}([0, 1])$.

Explainability: Scale-invariant coefficient of determination - explaining variance:

The scale-invariant R^2 equals

$$R^2(Y|\mathbf{X}) = \varrho_S(\psi(A)) \quad (7)$$

(see [63] for $d = 1$ and [20] for the case of arbitrary dimension) where $\varrho_S : \mathcal{C}^2 \rightarrow \mathbb{R}$ denotes Spearman's rank

²see, e.g., [65, 66], or more recently [21, 22, 29, 50, 54]

correlation coefficient³ given by

$$\varrho_S(C) := 12 \int_{[0,1]^2} C(s, t) \, d\lambda^2(s, t) - 3$$

and λ^2 the 2-dimensional Lebesgue measure on $\mathcal{B}([0, 1]^2)$.

From this perspective both initial $(d + 1)$ -dimensional problems - quantifying the scale-invariant extent of dependence $T(Y|\mathbf{X})$ of Y on $\mathbf{X}$, and calculating the scale-invariant coefficient of determination R^2 - can be reduced to the bivariate setting.

The bivariate copula $\psi(A)$ exhibits various interesting properties including the fact that the diagonal of $\psi(A)$ always exceeds the diagonal of Π ; see [20]:

1.2 Theorem.

1. $\psi(A)$ is invariant with respect to permutations and continuous, strictly monotone transformations of the coordinates of $\mathbf{X}$.
2. $\psi(A)$ satisfies the so-called information gain inequality along the diagonal, i.e., the more conditioning variables are involved, the larger the value of $\psi(A)$ along the diagonal. In particular,

$$\Pi(t, t) \leq \psi(A^{1,d+1})(t, t) \leq \dots \leq \psi(A^{1:k,d+1})(t, t) \leq \dots \leq \psi(A)(t, t) \quad (8)$$

for all $k \in \{1, \dots, d\}$ and all $t \in [0, 1]$; here $A^{1:k,d+1}$ denotes the marginal copula of A with respect to the coordinates $\{1, \dots, k\} \cup \{d + 1\}$.

Additionally, if $X_2, \dots, X_d, Y$ are conditionally independent given X_1 then the information gain inequality becomes an equality, i.e., $\psi(A) = \psi(A^{1,d+1})$.

The above-mentioned properties are inherited by T and the scale-invariant coefficient of determination R^2 :

Predictability: Measures of directed dependence of a random variable Y on a random vector $\mathbf{X}$.

$T(Y|\mathbf{X})$ is invariant with respect to permutations and continuous, strictly monotone transformations of $\mathbf{X}$, and T fulfills the *information gain inequality*, i.e.,

$$T(Y|X_1) \leq \dots \leq T(Y|X_1, \dots, X_k) \leq \dots \leq T(Y|X_1, \dots, X_d)$$

for all $k \in \{1, \dots, d\}$. Additionally, if $X_2, \dots, X_d, Y$ are conditionally independent given X_1 , then $T(Y|\mathbf{X}) = T(Y|X_1)$.

³also known as Spearman's ρ_S ; see, e.g., [18, 21, 50]

Explainability: Scale-invariant coefficient of determination - explaining variance:

$R^2(Y|\mathbf{X})$ is invariant with respect to permutations and continuous, strictly monotone transformations of $\mathbf{X}$, and $R^2(Y|\mathbf{X})$ fulfills the *information gain inequality*, i.e.,

$$R^2(Y|X_1) \leq \dots \leq R^2(Y|X_1, \dots, X_k) \leq \dots \leq R^2(Y|X_1, \dots, X_d)$$

for all $k \in \{1, \dots, d\}$. Additionally, if $X_2, \dots, X_d, Y$ are conditionally independent given X_1 , then $R^2(Y|\mathbf{X}) = R^2(Y|X_1)$.

Nearest neighbour based estimation: The transformation ψ allows the estimation of the quantities of interest to be performed in reduced dimension, i.e., bivariate.

For the bivariate copula $\psi(A)$ we propose an estimator whose form is reminiscent of the empirical copula (see, e.g., [16, 60]), but which is actually based on the graph-based so-called 'simple measure of conditional dependence' introduced by Azadkia and Chatterjee [3].

Let $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)$ be i.i.d. copies of $(\mathbf{X}, Y)$ with continuous marginals and underlying copula $A \in \mathcal{C}^{d+1}$. Since the univariate marginals are continuous ties only occur with probability 0. For each i , we denote by $N(i)$ the index j such that $\mathbf{X}_j$ is the nearest neighbour of $\mathbf{X}_i$ with respect to the Euclidean metric on $\mathbb{R}^d$. Since there may exist several nearest neighbours of $\mathbf{X}_i$ ties are broken at random.

For $(s, t) \in [0, 1]^2$, define

$$\hat{C}_n(s, t) := \frac{1}{n} \sum_{k=1}^n \mathbb{1}_{[0, s]}(G_n(Y_k)) \mathbb{1}_{[0, t]}(G_n(Y_{N(k)})) \quad (9)$$

where G_n denotes a renormalized version of the empirical distribution function of $Y_1, \dots, Y_n$, i.e.,

$$G_n(y) = \frac{1}{n+1} \sum_{k=1}^n \mathbb{1}_{(-\infty, y]}(Y_k)$$

By adapting the ideas used in [3], it is shown in [20] that $\hat{C}_n$ is a strongly consistent estimator for $\psi(A)$ (in full generality, i.e, without any smoothness assumptions). In particular, in the bivariate case ($d = 1$) the estimator $\hat{C}_n$ strongly consistently estimates the Markov product $A^t * A$.

Predictability: Measures of directed dependence of a random variable Y on a random vector $\mathbf{X}$.

Motivated by Equation (6), T can be estimated by plugging $\hat{C}_n$ into Spearman's footrule (see [22, 29, 50, 54]),

which results in

$$\hat{S}_n(Y|\mathbf{X}) := \varphi(\hat{C}_n) = 1 - \frac{3}{n(n+1)} \sum_{i=1}^n |R_i - R_{N(i)}| + \frac{3}{n(n+1)} \left(\frac{n(n+1)}{2} - \sum_{i=1}^n R_{N(i)} \right)$$

where R_i denotes the rank of Y_i among $Y_1, \dots, Y_n$, i.e., the number of j such that $Y_j \leq Y_i$; notice that there may exist more than one index i such that $\mathbf{X}_j$ is a nearest neighbour of $\mathbf{X}_i$ implying that $\sum_{i=1}^n R_{N(i)}$ may fail to equal $\frac{n(n+1)}{2}$. Consequently, $\hat{S}_n(Y|\mathbf{X})$ estimates $T(Y|\mathbf{X})$ strongly consistently. By construction, the estimator $\hat{S}_n$ is akin to the statistic $\hat{T}_n$ proposed by Akzadia & Chatterjee [3]

$$\begin{aligned} \hat{T}_n(Y|\mathbf{X}) &= \frac{\sum_{i=1}^n (n \min\{R_i, R_{N(i)}\} - L_i^2)}{\sum_{i=1}^n L_i (n - L_i)} \\ &= 1 - \frac{3}{n^2 - 1} \sum_{i=1}^n |R_i - R_{N(i)}| + \frac{3}{n^2 - 1} \left(\sum_{i=1}^n R_{N(i)} - \frac{n(n+1)}{2} \right) \end{aligned}$$

where L_i denotes the number of j such that $Y_j \geq Y_i$. It has been shown in [3] that $\hat{T}_n$ estimates T in a strongly consistent manner, and in [61] the authors showed asymptotic normality of $\sqrt{n}\hat{T}_n$ under independence and for some regularity conditions; for a comprehensive summary of properties for T_n we refer to Han [35] and the references therein.

Explainability: Scale-invariant coefficient of determination - explaining variance:

Motivated by Equation (7), the scale-invariant coefficient of determination R^2 can be estimated by plugging $\hat{C}_n$ into Spearman's ρ_S (see [54]), which results in

$$\hat{R}_n^2(V|\mathbf{U}) := \varrho_S(\hat{C}_n) = \frac{12}{n(n+1)^2} \sum_{i=1}^n R_i R_{N(i)} - 3 + \frac{12}{n(n+1)} \left(\frac{n(n+1)}{2} - \sum_{i=1}^n R_{N(i)} \right)$$

The estimator $\hat{R}_n^2(V|\mathbf{U})$ exhibits the typical Spearman's ρ_S structure (see [31, 54, 57, 59]), and estimates $R^2(V|\mathbf{U})$ strongly consistently.

Simulation study: For an illustration of the small and moderate sample performance of the estimator $\hat{C}_n$ we perform a simulation study and consider the dependence structure of an EFGM copula (see [12, Section 6.3]) with parameter $\alpha = 1$. We restrict ourselves to the case $d = 1$ in order to be able to interpret the results obtained also as estimator for the Markov product $A^t * A$. To test the performance of the estimator $\hat{C}_n$ in this setting, we generated samples of size $n \in \{20, 50, 100, 200, 500, 1.000, 5.000, 10.000\}$ and calculated $\hat{C}_n$. These steps were repeated $R = 1.000$ times. Figure 4 depicts the d_∞ -distance between our estimate and the true copula evaluated on a grid of size 50.

As can be seen from Figure 4, the copula estimate converges rather fast to the true copula showing an excellent asymptotic behaviour.

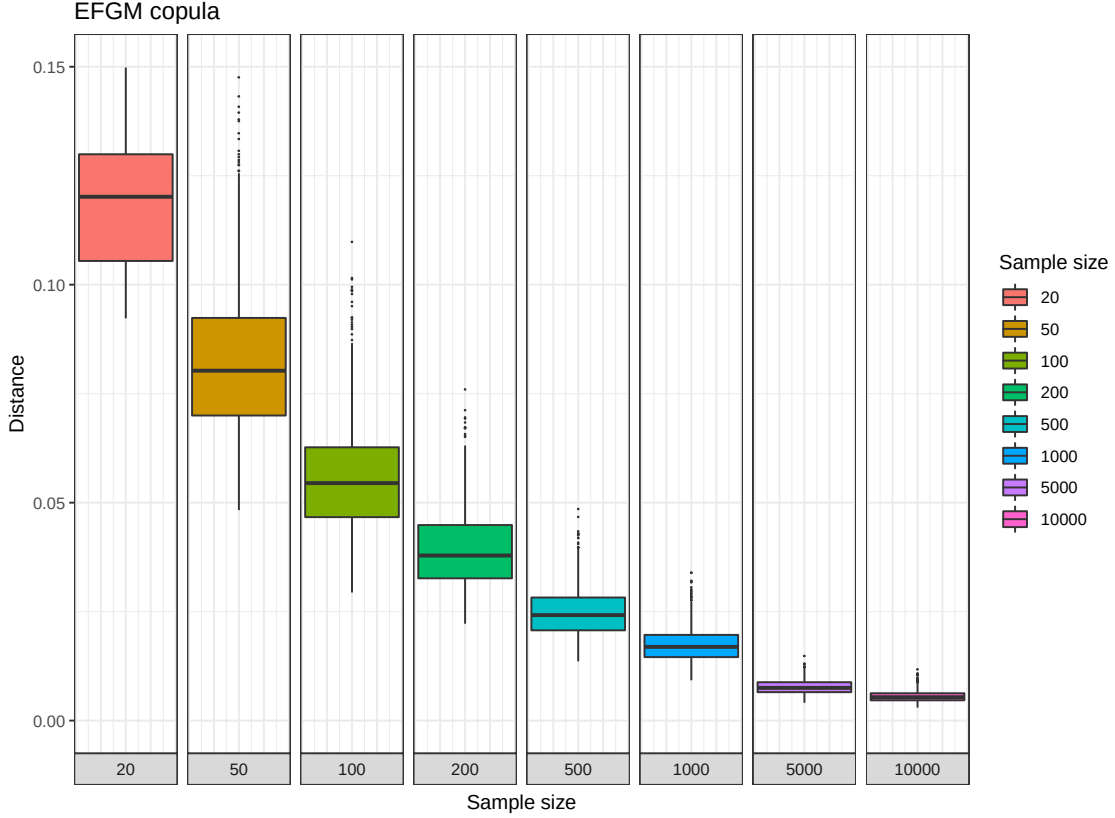


Figure 4: Boxplots summarizing the 1.000 obtained d_∞ -distances between our estimate and the true copula. Samples of size n are drawn from an EFGM copula A with parameter $\alpha = 1$.

Real data example - analysis of global climate data: We consider a data set of bioclimatic variables for $n = 1862$ locations homogeneously distributed over the global landmass from CHELSEA [44, 45] and want to analyse the influence of a set of thermal variables on annual precipitation.

We first focus on the case $d = 1$ and use the variable *Annual Mean Temperature* (AMT) to predict/explain the *Annual Precipitation* (AP). Figure 5 depicts the original dependence structure of AMT and AP (left panel) and the estimate $\hat{C}_n$ for $\psi(A)$ (right panel). The estimate $\hat{S}_n$ for T , i.e., for the scale-invariant extent of dependence of AP on AMT, equals 0.5603 and the estimate $\hat{R}_n^2$ for R^2 , i.e., for the scale-invariant coefficient of determination in this model, equals 0.6393.

In a second step, we extend the number of potential explanatory thermal variables to $d = 11^4$ to

⁴The considered thermal variables are *Annual Mean Temperature* (AMT), *Mean Diurnal Range* (MDR), *Isothermality* (IT), *Temperature Seasonality* (TS), *Max Temperature of Warmest Month* (MTWM), *Min Temperature of Coldest Month* (MTCM), *Temperature Annual Range* (TAR), *Mean Temperature of Wettest Quarter* (MTWeQ), *Mean Temperature of Driest Quarter* (MTDQ), *Mean Temperature of Coldest Quarter* (MTCQ), *Mean Temperature of Warmest Quarter* (MTWaQ).

predict/explain the *Annual Precipitation* (AP). Figure 6 depicts the estimate $\hat{C}_n$ for $\psi(A)$ where A describes the dependence structure of all variables involved. The estimate $\hat{S}_n$ for T , i.e., for the scale-invariant extent of dependence of AP on the considered 11 thermal variables, equals 0.6616 and the estimate $\hat{R}_n^2$ for R^2 , i.e., for the scale-invariant coefficient of determination in this model, equals 0.8105.

The increase in the estimated values for T and R^2 with increasing number of explanatory variables is in accordance with the information gain inequality.

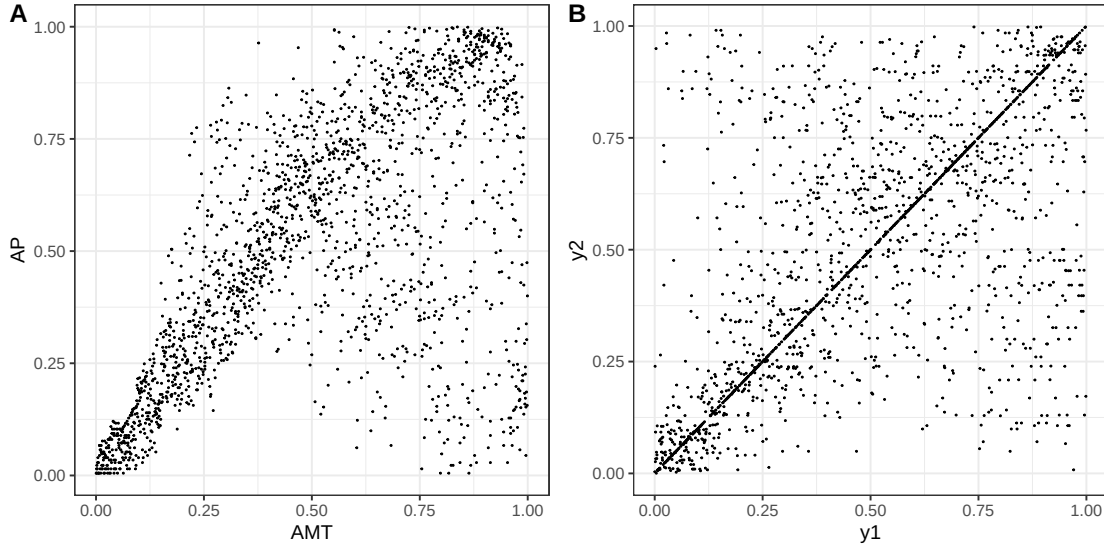


Figure 5: Original dependence structure of AMT and AP (left panel), together with the estimate $\hat{C}_n$ for $\psi(A)$ (right panel).

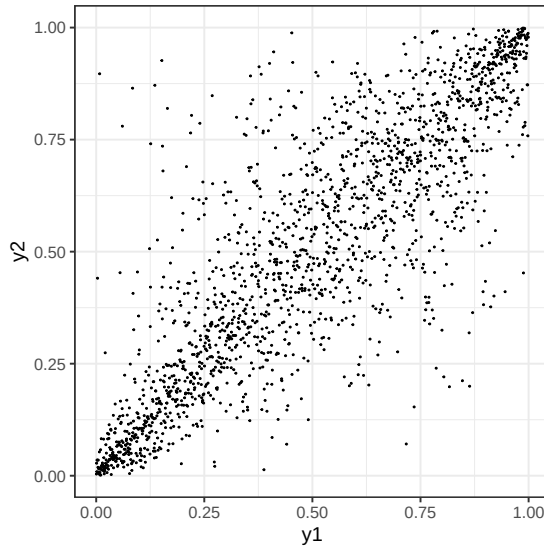


Figure 6: Estimate $\hat{C}_n$ for $\psi(A)$ where A reflects the 12-dimensional dependence structure of 11 thermal variables and AP.

1.2 Aims of the project and methods to realise them

The general aim of this project is to develop a comprehensive analytic copula-based framework for quantifying and estimating predictability and explainability of a single random variable Y using the information contained in a set of potential explanatory random variables $\mathbf{X} = (X_1, \dots, X_d)$ - including in particular the situation with d being large.

As outlined in Subsection 1.1, the transformation ψ , which enables a dimension reduction by converting the original $(d + 1)$ -dimensional problem into a 2-dimensional one, is key to this project. Therefore, we will put special emphasis on the following work packages:

- WP1 Development of estimation procedures based on checkerboard copulas and resampling methods to overcome drawbacks of the nearest neighbour estimator.
- WP2 Properties of the class Ψ and the development of new indices detecting complete dependence.
- WP3 Inference for the nearest neighbour based estimator $\hat{C}_n$.
- WP4 The case of arbitrary (not necessarily continuous) random variables.
- WP5 The implementation of the obtained methods in $\mathbb{R}$ to allow for broad applicability.

WP1 Checkerboard estimator and resampling methods

Although $0 \leq T \leq 1$ and $0 \leq R^2 \leq 1$, the estimators $\hat{T}_n$, $\hat{S}_n$ and $\hat{R}_n^2$ can attain large negative values which might be difficult to interpret for applicants (see Figure 7 (left panel) for an illustration). For the latter two indices, this is due to the fact that, in general, $\hat{C}_n$ itself is neither contained in Ψ nor is a copula at all. It hence may occur that its main diagonal lies below the main diagonal of Π (compare Eqn. (8)); Figure 7 (right panel) depicts the diagonal of $\hat{C}_n$ of a sample of size $n = 50$ compared to the main diagonal of Π .

The above drawback does no longer appear when using a checkerboard-based Markov kernel estimator for $\psi(A)$. The latter approach was developed by the project's postdoc candidate Florian Griessenberger and coauthors in [34] and [40] where it is used to estimate the index ζ_1 based on the metric D_1 (see, e.g., [71]). In contrast to $\hat{C}_n$, the checkerboard-based Markov kernel approach yields an estimator for $\psi(A)$ that is indeed a copula and is even an element of Ψ .

In addition, we will further investigate an estimator for $\psi(A)$ that is based on resampling methods; simulations indicate a promising well-behaviour of this estimation procedure.

The aim of WP1 is to develop estimation procedures that overcome the above-mentioned disadvantages of $\hat{C}_n$. For this purpose, we will apply the checkerboard-based Markov kernel approach and the resampling approach to our estimation problem and examine the asymptotic behaviour of the resulting estimators. Since

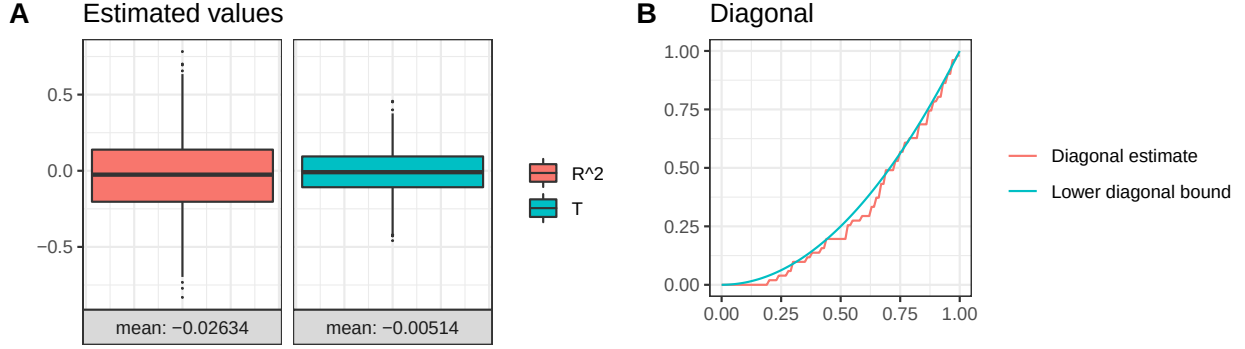


Figure 7: Boxplots summarizing the 1.000 obtained values for $\hat{R}_n^2$ and $\hat{S}_n$ drawn from samples of size $n = 50$ of two independent random variables uniformly distributed on $[0, 1]$ (left panel); lower diagonal bound of $\psi(A)$ compared to the diagonal of $\hat{C}_n$ of a sample of size $n = 50$ drawn from two independent random variables uniformly distributed on $[0, 1]$ (right panel).

Griessenberger has gathered plenty of know-how with checkerboard-based Markov kernel estimators, this part should be carried out by Fuchs and Griessenberger.

The specific tasks (T) and deliverables (D) of this work package are as follows:

- T 1.1: Development of a checkerboard-based Markov kernel estimator for $\psi(A)$ and investigation of its asymptotic properties.
- T 1.2: Development of an estimator for $\psi(A)$ based on resampling methods and investigation of its asymptotic behaviour.
- D 1.1: Publication(s) on inference for the checkerboard-based Markov kernel estimator for $\psi(A)$.
- D 1.2: Publication(s) on inference for the resampling-based estimator for $\psi(A)$.

WP2 The class Ψ and new indices detecting complete dependence

Although the copula $\psi(A)$ is bivariate, it contains the information whether Y is completely dependent on the d -dimensional random vector $\mathbf{X}$, and whether Y and $\mathbf{X}$ are independent. It has been shown in Subsection 1.1 (see also [20]) that the index T equals the dependence measure Spearman's footrule and that the scale-invariant coefficient of determination R^2 equals the popular measure of concordance Spearman's ρ_S both evaluating copula $\psi(A)$.

Applying $\psi(A)$ to other dependence measures such as Gini's γ or Kendall's τ (see, e.g., [13, 14, 18, 24]) leads to new indices with which complete dependence and, depending on the specific functional, also other facets of dependence can be detected: For instance, in the bivariate case (i.e., $d = 1$), evaluating $\psi(A)$ by Gini's γ coincides with the D_2 -metric (see, e.g., [8, 64, 71]) between the copula of (X, Y) and $(X, -Y)$, i.e.,

$$\begin{aligned}
\gamma_\psi(A) &:= \gamma(\psi(A)) = 2 \cdot D_2(A, \nu_2(A)) \\
&= 2 \cdot \int_{[0,1]} \int_{[0,1]} (K_A(u, [0, v]) - K_{\nu_2(A)}(u, [0, v]))^2 d\lambda(u)\lambda(v)
\end{aligned}$$

where $(\nu_2(A))(u, v) = u - A(u, 1 - v)$ is the copula of $(X, -Y)$. Consequently, γ_ψ is scale-invariant and satisfies properties (P.1) and (P.3), but fails to satisfy (P.2), in general. However, it turns out that $\gamma_\psi(A) = 0$ if and only if the dependence structures of (X, Y) and $(X, -Y)$ coincide, a kind of “lack of association” property⁵ that is fulfilled in case of independence. As a consequence, we obtain $\gamma_\psi(A) = 0$ whenever Y and X are independent.

The aim of this part is to systematically investigate the class $\Psi = \psi(\mathcal{C}^{d+1})$ of all bivariate copulas arising from ψ , and with the help of this class to develop new scale-invariant indices detecting complete dependence and also other facets of dependence. WP2 should be carried out by Fuchs and the doctoral student.

The specific tasks (T) and deliverables (D) of this work package are as follows:

- T 2.1: Investigation of properties of the class Ψ .
- T 2.2: Examination of existing measures of directed association whether based on Ψ .
- T 2.3: Development of new indices detecting complete dependence based on Ψ .
- D 2.1: Publication(s) on new measures of directed dependence.

WP3 Inference for the nearest neighbour based estimator $\hat{C}_n$

It has been shown in Fuchs [20] that the nearest neighbour based estimator $\hat{C}_n$ introduced in Eqn. (9) is a strongly consistent estimator for $\psi(A)$. We want to continue the study of the estimator’s asymptotic properties focusing initially on asymptotic normality.

Quite recently, Shi et al. [61] have shown asymptotic normality of the estimator $\sqrt{n} \hat{T}_n$ under independence and for some regularity conditions, and it seems possible that the methods used in [61] combined with the standard theory on the empirical copula process (see, e.g., [16, 31, 60]) are also applicable to the nearest neighbour based copula estimator $\hat{C}_n$ underlying $\hat{S}_n$ and $\hat{R}_n^2$. The index $\hat{T}_n$ can be computed in $O(n \log(n))$ time (see [62]) and we conjecture that this also holds for $\hat{C}_n$.

This part is one of the most analytically demanding one and requires solid mathematical knowledge in statistical inference; it should be carried out by Fuchs, Griessenberger and the doctoral student.

The specific tasks (T) and deliverables (D) of this work package are as follows:

- T 3.1: Inference for the nearest neighbour based estimator.
- D 3.1: Publication(s) on inference for the nearest neighbour based estimator.

⁵The term “lack of association” (indifference) has been introduced in an even stricter sense by Cifarelli et al. [6], more precisely, for bivariate distribution functions for which the dependence structures of (X, Y) , $(X, -Y)$ and even $(-X, Y)$ coincide.

WP4 Estimation and inference in the non-continuous case

As pointed out in Subsection 1.1, copulas capture all scale-invariant dependence structures between continuous random variables and from Sklar's theorem we can conclude that the underlying copula in Eqn. (4) is unique. However, if at least one of the marginal distribution functions fails to be continuous, the copula in Eqn. (4) remains unique only on a proper subset of the unit hypercube (see, e.g., [12, 54]) implying that there exist infinitely many copulas satisfying Eqn. (4). The non-uniqueness is crucial and does not allow the methods introduced above to be applied when dropping this assumption.

A popular approach to overcome the non-uniqueness problem consists in replacing the empirical copula by the empirical multilinear copula (see, e.g., [15, 28, 31, 32, 33, 54, 56]) also known as distributional transform or probability integral transform; its construction principle is exceptional as the non-observable and hence non-unique regions within the hypercube are interpolated linearly which ensures that no dependence is added to such regions.

The aim of WP4 is to extend the asymptotic theory on the empirical multilinear copula process to the nearest neighbour setting in order to be able to apply our estimation procedure also to non-continuous random variables and vectors. Considering that this part is also analytically highly demanding and requires solid mathematical knowledge in statistical inference it will be carried out by Fuchs, Griessenberger and the doctoral student.

The specific tasks (T) and deliverables (D) of this work package are as follows:

- T 4.1: Development of a nearest neighbour based estimator for $\psi(A)$ applicable to non-continuous data via the empirical multilinear copula approach.
- T 4.2: Investigation of the estimator's asymptotic properties.
- D 4.1: Publication(s) on the asymptotic behaviour of the nearest neighbour based estimator for $\psi(A)$ under broad conditions.

WP5 Software implementation, tutorials, and summary documents

The theoretical results that are derived within the above-mentioned work packages will be published in high rank peer-reviewed journals with a strong methodological focus. Additionally, in order to increase the visibility of the results and facilitate understanding and application of the proposed approaches, which are cross-disciplinary applicable, we will also implement the respective methods as R functions either in existing or in new R packages. Moreover, we will write corresponding tutorials, vignettes, and guidance papers that will be submitted to software journals.

Moreover, in this work package, a final summary report on the present project will be drafted, containing the key results and project outcomes as well as a discussion of open questions for future research.

The specific tasks (T) and deliverables (D) of this work package are as follows:

- T 5.1: Development of an R package for the methods considered in WP 1, WP 3 and WP 4.
- D 5.1: R package and corresponding tutorial/guidance for WP 1, WP 3 and WP 4.
- D 5.2: Final project report

1.3 Innovative aspects of the project

The innovative aspects of the proposed project can be grouped into two main categories:

1. Several key objects of the proposed project - this includes in particular the bivariate copula $\psi(A)$ capturing the information whether Y is completely dependent on the d -dimensional random vector $\mathbf{X}$, and whether Y and $\mathbf{X}$ are independent, but also the nearest neighbour based copula estimator $\hat{C}_n$ for $\psi(A)$ - have so far only been studied by Fuchs [20] and have otherwise not yet been addressed in the literature. To the best of our knowledge, these are truly innovative and very promising contributions with a broad range of applicability. Since they allow for a dimension reduction by converting the original $(d + 1)$ -dimensional problem into a 2-dimensional one, we expect our methods to be highly recognized in practice.
2. Rethinking existing and developing new indices being able to detect complete dependence (WP2) will, on the one hand, allow us to reinterpret and reclassify existing indices as functionals of $\psi(A)$ (as outlined in Section 1.1), and to construct new indices of complete dependence that focus on different facets of dependence relationships than those previously addressed in the literature. On the other hand, a fast, precise and efficient estimation of the above-mentioned existing and new indices is preferable when very large numbers of either continuous (WP1 & WP3) or non-continuous (WP4) explanatory variables are involved. This issue is of high practical relevance, and our research will help closing important knowledge gaps in the literature.

1.4 Timeline and project management

The total anticipated project duration is 36 months (3 years). According to the outline of the financial plan in Annex 2, the project leader Dr. Sebastian Fuchs applies for funding of one full-time (i.e., 40 h/week) postdoc and one full-time (i.e., 30 h/week) doctoral candidate both supervised by himself.

Timeline	1 st year	2 nd year	3 rd year
WP1 Checkerboard estimator and resampling methods T 1.1 Checkerboard estimator T 1.2 Resampling methods D 1.1 Publication(s) T 1.1 D 1.2 Publication(s) T 1.2			
WP2 The class Ψ and novel indices T 2.1 Properties of Ψ T 2.2 Existing indices based on Ψ T 2.3 New indices based on Ψ D 2.1 Publication(s) T 2.1 - T 2.3.			
WP3 Inference for the nearest neighbour estimator T 3.1 Inference D 3.1 Publication(s) T 3.1			
WP4 Estimation and inference in the non-continuous case T 4.1 Estimation T 4.2 Inference D 4.1 Publication(s) T 4.1 - T 4.2			
WP5 Software and tutorials T 5.1 Development of R package WP1, WP3 & WP4 D 5.1 R package and tutorial T 5.1 D 5.4 Final project report, open questions			

1.5 Collaborations

Continuing previous fruitful research collaborations, the project leader Dr. Fuchs plans to work together with the following national and international experts:

National collaboration partner:

- Univ.-Prof. Dr. Wolfgang Trutschnig (<https://www.trutschnig.net/>) is Professor for Stochastics at Paris-Lodron University Salzburg and Director of the Intelligent Data Analytics (IDA) Lab Salzburg (<https://ida-lab.sbg.ac.at/>). He will contribute his expertise in dependence modeling, copulas, nonpara-

metric statistics, and multivariate statistics, and he will bring in his experience with the checkerboard-based Markov kernel estimation procedure and with the measure of directed dependence ζ_1 he developed (see [34, 40, 71]). The involvement builds on a very intensive and efficient collaboration as illustrated by [26, 46, 47, 52, 53].

International collaboration partner:

- Prof. Dr. Fabrizio Durante (<https://sites.google.com/site/fbdurante/>) is Professor in Mathematical Methods of Economics, Finance and Actuarial Sciences at Università del Salento, Lecce (Italy). He is one of the most renowned experts in dependence modeling and author of the new standard reference work in copula theory [12]. Prof. Dr. Durante's involvement will be particularly related to WP2 and WP4 and will continue his long-standing and very successful collaboration with Dr. Fuchs, which has resulted in a number of peer-reviewed publications (e.g., [11, 21]).

1.6 Qualification of the investigators

Principal investigator (Dr. Sebastian Fuchs): Already Dr. Fuchs' doctoral thesis was about measuring dependence with copulas. After finishing his PhD at Technische Universität Dresden, Germany, he continued his studies in copula theory and, in several postdoc positions at TU Dresden, FU Bozen-Bolzano, TU Dortmund and University of Salzburg under the supervision of Klaus D. Schmidt, Fabrizio Durante, Joachim Kunert and Wolfgang Trutschnig he had the possibility to work on several novel approaches and research questions in the context of dependence modeling.

In particular, Dr. Fuchs focuses on transformations of copulas [9, 11, 19, 25, 46], on measuring and estimating dependence among an arbitrary number of random variables [1, 21, 22, 23, 47, 53], and the application of these methods in finance [2, 26, 52].

Quite recently, Dr. Fuchs developed the key objects for this proposal - the copula $\psi(A)$ and its nearest neighbour based copula estimator $\hat{C}_n$ [20] - which form the starting point for further investigation.

Postdoc candidate (Florian Griessenberger): F. Griessenberger (University of Salzburg) is a highly talented young mathematician who already wrote his master thesis on the measure of directed dependence ζ_1 that is based on the metric D_1 . He continued his study on that topic in his PhD thesis under the guidance of his supervisor Univ.-Prof. Dr. Wolfgang Trutschnig. During this time he published already four peer-reviewed articles including a version of ζ_1 for a single explanatory variable [40] and its extension to the case of an arbitrary number of explanatory variables [34]. These publications are complemented by the two R packages `qad` [39] and `qmd` [41] in which the developed methods were implemented (both available on CRAN). The checkerboard-based Markov kernel approach investigated in these articles will be crucial for WP1.

PhD student (N.N.): The offer for the PhD position will be officially announced and advertised via the common channels (like ST-Net). Ideally, the employee should have a solid background in measure theory, dependence modeling and nonparametric rank-based methods, because he/she will mainly work on WP2, WP3 and WP4. The PhD thesis will focus on quantifying and estimating directed dependence.

Impact on the career of the principal investigator: This project would have great impact on the career of the principal investigator. Being a Senior Scientist, it is the natural next step to build a small but productive research group and third party funding is the only way that allows Dr. Fuchs to employ PhD students or postdocs and thus gain the experience necessary to successfully apply for a full professorship in the future. This project will help to extend his expertise in the field of dependence modeling and to expand and deepen his contacts to experts all over the world working in the field.

Impact on the career of the postdoc: For F. Griessenberger working on the project will be most significant. After his very productive time as doctoral student this position will allow him to work on challenging problems in an active team and thus establish himself in the very active and strong community working in dependence modeling. He will be able to strengthen and broaden his research profile. This will certainly enable him to take the next steps of his promising academic career.

Impact on the career of the PhD student: The PhD student will get in contact with challenging, exciting, highly topical research questions, and this position will open the door for him/her to the very active and strong community working in dependence modeling. He/she will certainly not only write a fine PhD thesis but also start to build his/her name in the community.

1.7 Ethical, safety-related, and regulatory aspects

In the proposed project, we will only re-analyze data that has been already collected - Therefore, no ethical or safety-related issues will arise in this respect. The R codes and the corresponding software implementations will be open source and publicly available. The same applies to the fully anonymized versions of the datasets, provided that we obtain permission by the respective data owners, and that all legal and ethical requirements are met.

Apart from that, no further ethical, safety-related or regulatory issues are relevant to the proposed project.

1.8 Gender- and sex-related aspects

Due to the fact that the main focus of the proposed project is on developing novel methods for detecting complete dependence, sex- and gender-related aspects might be relevant only to a limited extent. However, for

illustrating the application of our proposed methods in our publications, we will preferably use datasets that allow for addressing such issues, if possible. Apart from that, the statistical techniques under consideration and the corresponding software enable researchers to conduct analyses addressing sex- and gender-related aspects. Analogous arguments apply to diversity aspects in general.

At the project group level, there is a clear male predominance. However, we plan to compensate for that by offering the funded doctoral position preferably to a non-male researcher, given that the person meets the essential requirements for the position.

Annex 1: References

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Annex 2: Financial aspects

Details on the research institution of the applicant.

The project will take place at the Department for Artificial Intelligence and Human Interfaces (AIHI) of the Paris Lodron University Salzburg (PLUS) that will provide a perfectly equipped and completely functioning scientific working environment. The complete infrastructure of the AIHI department will be at our disposition. The department will provide all project members with office space including a personal computer.

The project leader Dr. Sebastian Fuchs has a full-time Senior Scientist (postdoc) position at the AIHI department. He is also head of the team “Dependence Modeling” that is embedded in the Lab for Intelligent Data Analytics (IDA Lab) Salzburg.

The Department for Artificial Intelligence and Human Interfaces (AIHI) (<https://www.plus.ac.at/aihi/>) was founded 2022 as part of a larger restructuring process at the PLUS and the establishment of the Faculty of Digital and Analytical Sciences (DAS). The AIHI department consolidates competences in the areas of statistics/stochastics, data science, machine learning, visual computing and human-computer interaction. The statistics/stochastics and data science group at the department consists of two full professors, one Senior Scientist and 6 PhD students. Univ.-Prof. Dr. Arne Bathke is a leading expert in nonparametric and multivariate statistics, and Univ.-Prof. Dr. Wolfgang Trutschnig is a leading researcher in the field of dependence modeling and director of IDA Lab Salzburg. The project will greatly profit from their expertise and that of their very promising PhD students as well.

Sebastian Fuchs’ team “Dependence Modeling” (<https://ida-lab.sbg.ac.at/teams/dependence-modeling/>) is embedded in the Lab for Intelligent Data Analytics Salzburg (IDA Lab) funded by the federal state of Salzburg (WISS 2025, project IDs 20102-F1901166-KZP and 20204-WISS/225/197-2019, <https://ida-lab.sbg.ac.at/>). The IDA Lab project aims at establishing an overarching center for statistics, data science, and machine learning, where all relevant research institutions in Salzburg are involved. The IDA Lab activities are focused on basic methodological research with an underlying real-life motivation, and various projects are conducted jointly with partners from academia and industry. Currently, the research team of IDA Lab Salzburg consists of three full-time professors, five postdocs, eight PhD students, and several student assistants. At IDA Lab there is also experienced personnel available that may assist in project management and administrative issues. Moreover, the IDA Lab framework also greatly facilitates involving further experts that might provide relevant contributions to the proposed project: For example, Univ.-Prof. Dr. Roland Kwitt (PLUS), professor for Machine Learning, and Univ.-Prof. Dr. Christian Borgelt (PLUS), professor for Data Science, are both vice-directors of IDA Lab.

Information on the funding requested.

For the whole project period of 36 months we request

- one post doctoral position (project's postdoc candidate Florian Griessenberger) for 40 hours/week (100%), and
- one doctoral candidate position for 30 hours/week (75%).

The offer for the doctoral position will be officially announced and advertised within the national and international scientific community (e.g., via ST-NET). The project team will set up a two-stage process, asking for job applications in writing at first, creating a shortlist of candidates and interviewing these eligible applicants at the second stage. In particular, gender and diversity aspects will be adequately considered in the selection process.

Postdoc candidate (Florian Griessenberger, 36 month, 100%): With the experience from two publications on the measure of directed dependence ζ_1 and its estimation using the checkerboard Markov kernel approach (see [34, 40]), complemented by the implementation of the developed methods as R packages (see [39, 41]), F. Griessenberger is well qualified to tackle the problems described in WP1, WP3 and WP4. A variety of sophisticated tools from statistics and probability theory have to be expertly mastered in order to successfully carry out this research. The project will strongly benefit from his expertise and involvement.

PhD student (N.N., 36 month, 75%): The PhD candidate should tackle the tasks mentioned in WP2 and WP3 and, to some extent, those described in WP4. Since WP2 and WP3 have both, parts where we can guarantee success as well as parts where the candidate can show his/her abilities, it seems well suited for the beginning of a PhD project. WP4 is more demanding, requires solid mathematical knowledge of statistical inference and will be therefore addressed at a later stage of the project. This enables the candidate to quickly achieve first results and then move on and become an expert in the field by the time he/she finishes his/her PhD.

Total costs including general costs calculated by the FWF

	1 st year	2 nd year	3 rd year	Total
Personnel				
Post doctoral candidate 40 h/week	€ 72.030,00	€ 72.030,00	€ 72.030,00	
Doctoral candidate 30 h/week	€ 40.860,00	€ 40.860,00	€ 40.860,00	
Interim total	€ 112.890,00	€ 112.890,00	€ 112.890,00	€ 338.670,00
General costs (5 %)				€ 16.933,50
Total sum				€ 355.603,50

Annex 3: CVs and description of previous research achievements

Sebastian Fuchs

Short Bio

Sebastian Fuchs studied mathematics with minor in economics at Technical University Dresden (TU Dresden), where he passed his diploma degree in 2010 and completed his PhD (grade: summa cum laude, with distinction) in 2015. After substituting the professorship for Applied Stochastics at TU Dresden (2016-2017), he worked as postdoctoral researcher at the Faculty of Economics and Management at Free University of Bozen-Bolzano in Italy (2017-2018), and at the Faculty of Statistics at Technical University Dortmund in Germany (2018-2019). In 2019, he started as Senior Scientist at the Paris-Lodron University Salzburg (PLUS), Austria, where he is part of the research group of Wolfgang Trutschnig. Since 2020 Sebastian Fuchs is leading the team ‘Dependence Modeling’ at Salzburg Lab for Intelligent Data Analytics (IDA Lab).

Equipped with a strong background in actuarial and financial mathematics, Sebastian Fuchs mainly works on dependence modeling with copulas, being particularly interested in aspects of dependence modeling that go beyond the bivariate case. Focusing his research on copulas during and after his PhD he became an expert in multivariate dependence modeling and is well known for his research on measuring and modeling multivariate association. Sebastian Fuchs is editorial advisory board member for ‘Dependence Modeling’ and has published more than 20 papers in international journals - mainly mathematics/statistics but also interdisciplinary ones with colleagues from biology, economics, and finance - [List of Publications](#). His team Dependence Modeling (at IDA Lab Salzburg), which currently consists of three PhD students and three student assistants, actively collaborates with companies and solves real-life Data Science problems applying and conducting basic research in Statistics and Machine Learning.

Current Position

since 10/2019 **Senior Scientist**, *Department for Artificial Intelligence and Human Interfaces*,
Paris-Lodron University Salzburg, Austria.

since 10/2020 **Head of Team Dependence Modeling at Lab for Intelligent Data Analytics**
(IDA Lab Salzburg), *Paris-Lodron University Salzburg, Austria.*

Previous Positions

- 04/2018 – 09/2019 **Postdoc Researcher**, *Faculty of Statistics*, TU Dortmund, Dortmund, Germany.
- 04/2017 – 03/2018 **Postdoc Researcher**, *Faculty of Economics and Management*, Free University of Bozen-Bolzano, Bozen-Bolzano, Italy.
- 10/2016 – 03/2017 **Substitute of the Professorship for Applied Stochastics**, *Institute of Mathematical Stochastics*, TU Dresden, Dresden, Germany.
- 03/2010 – 09/2016 **Research Assistant / PostDoc Researcher**, *Institute of Mathematical Stochastics*, TU Dresden, Dresden, Germany.
- Paternal leave (twice, two months each time).

Education

2015 **PhD**, *Technical University Dresden*, Germany.

- Degree: Dr. rer. nat. (November 2015)
- Grade: Summa cum laude (with distinction)
- Title of dissertation: Transformation of Copulas and Measures of Concordance

2010 **Diplom/MSc**, *Technical University Dresden*, Germany.

- Degree: Dipl.-Math.
- Grade: Sehr gut (very good)
- Special Topics: Probability Theory and Statistics, Actuarial Mathematics

Ten Most Important Scientific Publications

Mroz T., S. Fuchs, and W. Trutschnig (2021). How simplifying and flexible is the simplifying assumption in pair-copula constructions - analytic answers in dimension three and a glimpse beyond. *Electron. J. Stat.* 15(1), 1951–1992. doi 10.1214/21-EJS1832

Kasper T., S. Fuchs, and W. Trutschnig (2021). On weak conditional convergence of bivariate Archimedean and Extreme Value copulas, and consequences to nonparametric estimation. *Bernoulli* 27(4), 2217–2240. doi 10.3150/20-BEJ1306

Fuchs S., F.M.L. Di Lascio, and F. Durante (2021). Dissimilarity functions for rank-based hierarchical clustering of continuous variables. *Comput. Statist. Data Anal.* 159, Article ID 107201, 26 pages. doi 10.1016/j.csda.2021.107201

Ahn J.Y., S. Fuchs, and R. Oh (2021). A copula transformation in multivariate mixed discrete-continuous models. *Fuzzy Set. Syst.* 415, 54–75. doi 10.1016/j.fss.2020.11.008

Fuchs S. and W. Trutschnig (2020). On quantile based co-risk measures and their estimation. *Depend. Model.* 8, 396–416. doi 10.1515/demo-2020-0021

Ahn J.Y. and S. Fuchs (2020). On minimal copulas under the concordance order. *J. Optim. Theory Appl.* 184, 762–780. doi 10.1007/s10957-019-01618-4

Durante F. and S. Fuchs (2019). Reflection invariant copulas. *Fuzzy Set. Syst.* 354(1), 63–73. doi 10.1016/j.fss.2018.02.004

Fuchs S., Y. McCord, and K.D. Schmidt (2018). Characterizations of copulas attaining the bounds of multivariate Kendall's tau. *J. Optim. Theory Appl.* 178(2), 424–438. doi 10.1007/s10957-018-1285-6

Dietz M., S. Fuchs, and K.D. Schmidt (2016). On order statistics and their copulas. *Statist. Probab. Lett.* 117, 165–172. doi 10.1016/j.spl.2016.05.020

Fuchs S. (2014). Consistent loss prediction for a portfolio and its subportfolios. *Scand. Actuar. J.* 2014(6), 561–581. doi 10.1080/03461238.2012.749508

Main Areas of Research

Copulas and Dependence Modeling

Measures of Association

Multivariate and Nonparametric Statistics

Quantitative Risk Management

Actuarial and Financial Mathematics

Data Analytics and Feature Selection.

Additional Important Scientific Achievements

- 21 publications in peer-reviewed journals - [List of Publications](#) - [ORCID: 0000-0002-9317-6878](#).
- Head of Team Dependence Modeling at Lab for Intelligent Data Analytics (IDA Lab) Salzburg which currently consists of three PhD students and three student assistants - [IDA-Lab Team DeMo](#).
- PI of research project (2021): Financial Fitness Score (FinFit), in collaboration with *Bausparkasse Wüstenrot AG*.
- Assistant Editor and Editorial Advisory Board Member for *Dependence Modeling*.
- Reviewer for more than 15 international journals in mathematics/statistics.
- Co-Advisor for Athinan Sutchada, a Ph.D. student at the School of Science at Walailak University, Thailand (since 2021).
- Member of Ph.D. jury for Vojtech Kika, a Ph.D. student at KU Leuven, Belgium, and Charles University, Prague, Czech Republic (2021).

Florian Griessenberger

Current Position

since 11/2021 **Research assistant (Predoc)**, *Department for Artificial Intelligence and Human Interfaces*, Paris-Lodron University Salzburg, Austria, (20 hours/week).

Previous Positions

11/2018 - 10/2021 **Research assistant (Predoc)**, *START project: Successional Generation of Functional Multidiversity*, Paris-Lodron University Salzburg, Austria, (30 hours/week).

10/2016 - 06/2018 **Teaching assistant**, *Department of Mathematics and Department of Psychology*, Paris-Lodron University Salzburg, Austria, (10 hours/week).

Education

since 11/2018 **PhD in Mathematics**, *University of Salzburg*, Austria.

08/2016 – 10/2018 **Master of Science in Mathematics**, *University of Salzburg*, Austria.

10/2012 – 08/2016 **Bachelor of Science in Mathematics**, *University of Salzburg*, Austria.

10/2012 – 07/2015 **Bachelor of Science in Psychology**, *University of Salzburg*, Austria.

Most Important Scientific Publications

Griessenberger F., J. Fernández-Sánchez, and W. Trutschnig (2021). Some properties of double shuffles of bivariate copulas and (extreme) copulas invariant with respect to Luroth double shuffles. to appear in *Fuzzy Set. Syst.* doi: 10.1016/j.fss.2021.02.014

Junker R.R., F. Griessenberger, and W. Trutschnig (2021). Estimating scale-invariant directed dependence of bivariate distributions. *Comput. Statist. Data Anal.* 153, Article ID 107058, 22 pages. doi 10.1016/j.csda.2020.107058

Griessenberger F. and W. Trutschnig (2020). Estimating asymmetric dependence via empirical checkerboard copulas. In A. Pollice, N. Salvati, and S. Spagnolo (Eds.), *Book of Short Papers SIS 2020*, pp. 304–309.

Junker R.R., M. Hanusch, X. He, V. Ruiz-Hernandez, J.C. Otto, S. Kraushaar, K. Bauch, F. Griessenberger, L.-M. Ohler, and W. Trutschnig (2020). Ödenwinkel: An alpine platform for observational and experimental research on the emergence of multidiversity and ecosystem complexity. *Web Ecol.* 20, 95–106. doi 10.5194/we-20-95-2020

Griessenberger F., R.R. Junker, and W. Trutschnig: On a multivariate copula-based dependence measure and its estimation. Available at <https://arxiv.org/abs/2109.12883>.

Main Areas of Research

Copulas and Dependence Modeling

Measures of Association

Multivariate and Nonparametric Statistics

Data Analytics and Feature Selection

Additional Important Scientific Achievements

- 3 publications in peer-reviewed journals - [List of Publications](#) - [ORCID: 0000-0003-0925-5416](#).
- Honours and awards: Winner of the Hans-Stegbuchner-Preis, University of Salzburg, 01/2019.
- Software:
 - Junker R.R., F. Griessenberger, and W. Trutschnig (2019). `qad`: Quantification of Asymmetric Dependence. R package version 1.0.1. Available on CRAN.
 - Junker R.R., F. Griessenberger, and W. Trutschnig (2021). `qmd`: Quantification of Multivariate Dependence. R package version 1.0.0. Available on CRAN.
- Talks
 - Quantifying And Estimating Asymmetric Dependence, at the 6th Joint PhD Colloquium in Statistics and Stochastics of the Center for Statistics in Oldenburg and Bremen (ZeSOB). 09/2021.
 - Quantifying And Estimating Asymmetric Dependence, at the 12th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2019), London, UK. 12/2019.
 - QAD. An R-package for Quantifying (Asymmetric) Dependence, at the 10th International Workshop on Simulation and Statistics, Salzburg, Austria. 09/2019.
 - QAD. An R-package for Quantifying Asymmetric Dependence, at the Symposium: Ecology – Geomorphology – Statistics, Salzburg, Austria. 03/2019.

Annex 4: Collaboration letters

Univ.-Prof. Dr. Wolfgang Trutschnig
Professor for Stochastics
Director IDA Lab Salzburg
Department for Artificial Intelligence and
Human Interfaces
University of Salzburg
Hellbrunnerstrasse 34
A - 5020 Salzburg
✉ wolfgang.trutschnig@sbg.ac.at
🏠 www.trutschnig.net, ida-lab.sbg.ac.at

Salzburg, 2022-02-12

To whom it may concern,

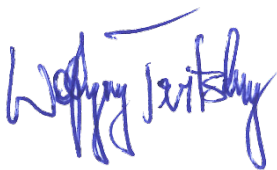
It is my pleasure to collaborate with Dr. Sebastian Fuchs within his project **ReDim: Quantifying dependence via dimension reduction**.

Since 2019 Dr. Fuchs is senior scientist in my working group on dependence modeling (stochastics) at the Paris Lodron University Salzburg (PLUS) and I am happy to say that our collaboration during the past 2.5 years has been a quite productive one (as reflected by the papers [1] and [2] below).

After having read his ReDim proposal in detail I am convinced that, firstly, the tackled problems are thrilling and – if solved in a definitive manner – exhibit a very broad range of applicability throughout all quantitatively working disciplines (where quantities like R^2 are omnipresent). And, secondly, building on the tools we have derived and studied over the past years (the checkerboard estimator being only one prominent example with a surprisingly good performance) I do think that it is possible to solve all sketched problems (as well as additional ones which will naturally come up as by-products).

I do think that Dr. Fuchs' proposal deserves full funding, expect various new methods as well publications in basic research and some application papers to emerge from the project, and look forward to continuing my fruitful collaboration with him and his future project members.

Yours sincerely



Wolfgang Trutschnig

[1] T. Kasper, S. Fuchs, W. Trutschnig: On weak conditional convergence of bivariate Archimedean and Extreme Value copulas, and consequences to nonparametric estimation, *Bernoulli* 27(4), 2217-2240 (2021)

[2] T. Mroz, S. Fuchs, W. Trutschnig: How simplifying and flexible is the simplifying assumption in pair-copula constructions - analytic answers in dimension three and a glimpse beyond, *Electronic Journal of Statistics* 15, 1951-1992 (2021)



DIPARTIMENTO DI SCIENZE DELL'ECONOMIA

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TO WHOM IT MAY CONCERN,

it is my pleasure to collaborate with Dr. Sebastian Fuchs within his project

ReDim: Quantifying dependence via dimension reduction.

I started my scientific contacts with Dr. Fuchs in 2015 when he was doctoral student at the Technical University of Dresden. At that time, he was working on his PhD dissertation about the interplay between measures of concordance and copulas.

Since then, Dr. Fuchs continues working in the field of stochastic dependence and copulas by giving a profound and rigorous mathematical framework to various statistical methods and procedures relevant for applications. Nowadays, his studies have been published in several internationally renowned journals and he is a well-recognized scientist in the international community of dependence models.

I read carefully the ReDim project proposal and I am convinced that it contains various novel ideas in the field of association measures. In particular, it aims at quantifying predictability and explainability of a single random object by using a combination of techniques from classical statistics to machine learning methodologies (like nearest-neighbour algorithms) that are, in my opinion, adequate to solve the proposed problems.

My collaboration within the project, together with the research team that will be built by Dr. Fuchs, will be particularly related to WP2 and WP4. I find that this latter working package is particularly challenging because copula techniques have been mainly developed for continuous data and an extension to mixed type data is needed to enhance different kind of applications.

Overall, I think that the project proposal is well structured, promotes innovations, and will have an impact in all the applied fields where relationships among different uncertain phenomena must be explained and predictions are needed.

Moreover, I am convinced that Dr Fuchs has the potential to lead the research team and to pursue the expected output. Thus, any kind of support given to this research proposal will guarantee benefits to the scientific research and the overall society.

Yours sincerely,

Fabrizio Durante